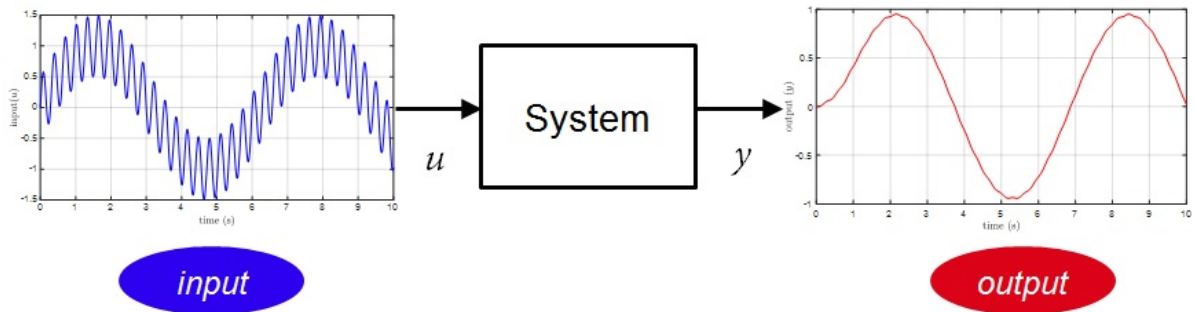


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QUIZ II - LAPLACE TRANSFORM

1. Consider the system



The transfer function of the system is given by one of the following options - which one?

(A) $H(s) = 10 \quad Y(s) = 10 X(s) \rightarrow y(t) = 10 x(t)$

(B) $H(s) = \frac{1}{(s-3)^2(s+5)}$ ← unstable

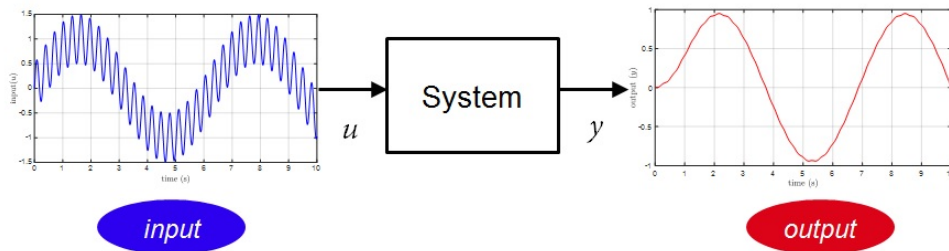
↑ 2 poles in the RHP

(C) $H(s) = \frac{125}{(s+5)^3}$ ← stable

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QUIZ II - LAPLACE TRANSFORM

2. Consider the system



The input signal is $x(t) = \sin(t) + 0.5 \sin(20t)$.

The transfer function is $H(s) = \frac{125}{(s+5)^3}$.

What is the s -domain output signal?

(A) $Y(s) = \frac{125}{(s+5)^3} \left(\frac{1}{s^2+1} + \frac{10}{s^2+400} \right)$

(B) $Y(s) = \frac{125}{(s+5)^3} \left(\frac{1}{s^2+1} + \frac{10}{s^2+20} \right)$

(C) $Y(s) = \frac{125}{(s+5)^3} \left(\frac{1}{s^2+1} + \frac{20}{s^2+400} \right)$

(D) $Y(s) = \frac{125}{(s+5)^3} \left(\frac{s}{s^2+1} + \frac{0.5s}{s^2+400} \right)$ ✓ from the Laplace Transform table

$$Y(s) = H(s)X(s) \quad X(s) = \mathcal{L}(\sin(t)) + 0.5 \mathcal{L}(\sin(20t))$$

$$= \frac{1}{s^2+1} + 0.5 \frac{20}{s^2+(20)^2}$$

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QUIZ II - LAPLACE TRANSFORM

3. A system has transfer function

$$H(s) = \frac{1}{(s+1)(s+2)}$$

$\swarrow s = -1 \quad \swarrow s = -2$

and input

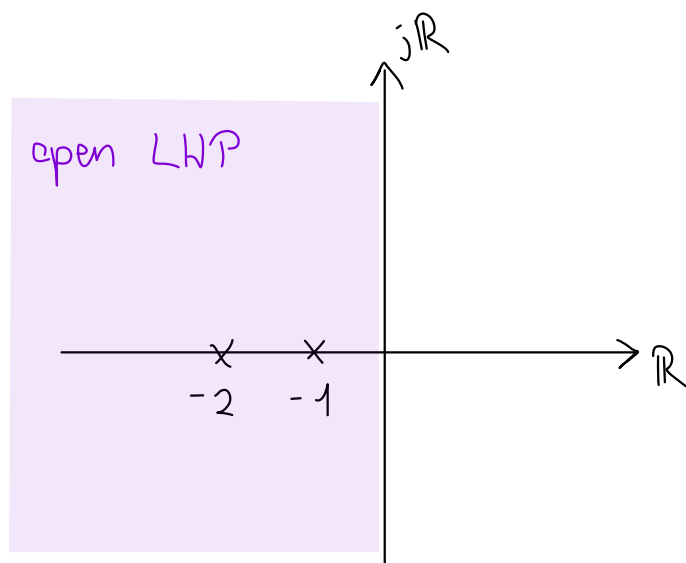
$$X(s) = \frac{1}{s-1}.$$

Is the system stable?

(A) It is stable for $s > 1$

(B) No

(C) Yes



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QUIZ II - LAPLACE TRANSFORM

4. A system has transfer function

$$H(s) = \frac{1}{s^2 LC + sRC + 1}.$$

If $L = 1$, $C = 1$, pick R such that the system has a double pole at $s = -1$.

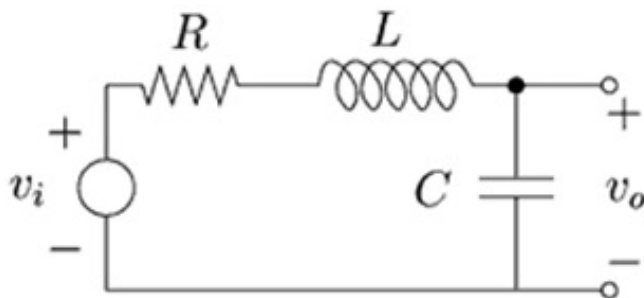
(A) $R = 4$

$$\begin{aligned} (s^2 + R s + 1) &= (s + 1)^2 \\ &= s^2 + 2s + 1 \end{aligned}$$

(B) $R = 2$

$$\hookrightarrow R = 2$$

(C) $R = 0$



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QUIZ II - LAPLACE TRANSFORM

5. Consider the system

$$K(s) = k \frac{1}{s-1}$$

1 pole at $s=1$
 $\hookrightarrow K(s)$ is unstable

For which values of $k \neq 0$ does the system have a bounded output (given a bounded input)? for any value of k ?

(A) $k > 1$

(B) $k < 1$

(C) None

(D) Some other value of k